Civilisational Layer Theory

A Thermodynamic Framework for Civilisational Transitions, Human–AI Symbiosis, and the Limits of Intelligence

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Abstract

Every major civilisational transition follows the same structural pattern: it resolves one form of scarcity, creates a new one, and splits the population into two stable attractors — a stronger elite and a larger atrophied majority. This paper presents a thermodynamically grounded framework, built from three formal streams (F^-, F^+, F^{auto}) and the agency operator F^Φ, that unifies all such transitions into a single theory of ontological layers. We trace the pattern from the earliest known example of inter-species symbiosis — the human–dog co-evolutionary bond beginning approximately 40,000 years ago — through the Industrial, Information, and current AI revolutions, to a formal prediction of a convergence point at $T^* \approx 2043$ CE. All claims carry explicit epistemic tags: **[D]** (derived), **[C]** (conjecture), **[P]** (falsifiable prediction), **[X]** (beyond current reach). Physics is treated as immutable; statistics as negotiable.

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1 The Deepest Precedent: Human–Dog Co-evolution

The central argument of this paper is that *human–AI symbiosis is not new.* It is a repetition, at a new layer, of a pattern that is at least 40,000 years old. The first and most revealing instance of that pattern is the relationship between *Homo sapiens* and the grey wolf (*Canis lupus*).

1.1 The Co-evolutionary Bond

Genetic and archaeological evidence places the domestication of wolves between 40,000 and 15,000 years ago — the most conservative estimates cluster around 15,000 BCE in Eastern Eurasia, while behavioural and mitochondrial evidence pushes the process back further, possibly to a mutual self-selection dynamic rather than deliberate human intent [1, 2, 3].

What makes this event structurally significant is not its antiquity but its mechanism. Neither humans nor wolves *designed* the partnership. The wolf that tolerated human proximity gained access to caloric resources (refuse, hunt residue) it could not otherwise reliably obtain. The human group that tolerated wolves nearby gained an olfactory and auditory detection system of a quality no human nervous system could replicate. The boundary between *tool* and *partner* was, from the beginning, undefined — and this ambiguity was not a problem to be solved but a productive feature of the arrangement.

Structural Signature of Productive Symbiosis

A symbiosis is *productively structured* when: (i) each component provides a capability the other cannot synthesise internally; (ii) the combined system produces outputs neither component can produce alone; and (iii) the boundary between components is permeable, not fixed.

All three conditions were satisfied by the human–wolf bond from its earliest detectable phase. The wolf's olfactory range is approximately 10,000 times that of a human — providing an early-warning and tracking system that extended human agency across spatial scales inaccessible to unaided perception. The human, in return, provided *directional intentionality*: the capacity to choose targets, coordinate pursuit strategies, and preserve kill surplus across time. Neither species had this combined capability. The hybrid system did.

1.2 The Bifurcation That Followed

The domestication event did not uniformly affect all human groups. The populations that integrated dogs into their hunting and herding systems gained a decisive differential advantage in caloric yield, territorial range, and defensive capacity. Those that did not remained within the limits of unaided human sensing and social coordination. This is not a moral judgement; it is a structural observation: *the introduction of a new layer created a bifurcation in human capability*, and the gap between the two branches was not linear — it was multiplicative.

This is the first recorded instance of what we will formalise in Section 4 as the **Bifurcation Law**.

1.3 The Atrophy Signal

The bond also carried the first recorded signal of what we will later call *Agency Capital erosion*. Populations that had integrated dogs for olfactory tracking showed measurable reduction in the olfactory cortex relative to isolated populations [4]. The capability that had been offloaded did not remain in reserve — it declined. The organism allocated fewer resources to what the environment no longer required of it. This is not pathology. It is efficient thermodynamics. The pathology arises only when the external system is suddenly removed and the internal capacity has already been reabsorbed.

This mechanism — offloading produces atrophy, atrophy is invisible until the tool disappears — recurs without exception in every subsequent layer transition.

1.4 Further Historical Instances of Hybrid Systems

Table 1: Recurrent instances of human–tool symbiosis before the AI layer

Hybrid System	Period	Capability Gained	Capability Atrophied
Human + Dog	∼40,000–15,000 BCE	Olfactory range, herding, defence	Olfactory cortex density
Human + Horse	∼3,500 BCE	Mobility, load transport, warfare range	Long-distance walking endurance
Human + Writing	∼3,200 BCE	Externalised memory, coordination at scale	Oral memory depth (epic tradition collapses)
Human + Agriculture	∼10,000 BCE	Caloric surplus, sedentary complexity	Foraging skill, skeletal robustness
Human + Printing Press	1440 CE	Information diffusion, individual reading	Oral transmission chains, scribal guilds
Human + Calculator	1970s CE	Arithmetic speed at scale	Mental arithmetic fluency
Human + GPS	2000 CE	Navigation without spatial knowledge	Hippocampal spatial mapping (measurable) [5]
Human + LLM	2023 CE	Language synthesis, knowledge retrieval	Agency over epistemic functions (onset)

The pattern in Table 1 is not accidental. It reflects a structural property of any system that offloads a function to an external substrate: the internal implementation of that function is down-regulated by metabolic optimisation. The question is never whether atrophy occurs — it always does. The question is whether the offloaded function can be recovered when needed, and whether the system has gained capabilities that compensate.

2 The Three-Stream Framework

To give the historical pattern a precise formal structure, we require a language capable of describing any complex system — physical, biological, cognitive, or civili-

sational — through the same set of operators. We adopt the following foundational decomposition.

2.1 The Three Streams

Definition 2.1 (Three-Stream Decomposition)**.** *Any complex system S is characterised by three orthogonal dynamic streams:*

$$F^{-}: \quad \left.\frac{dS}{dt}\right|_{entropic} \geq 0 \qquad \textit{thermodynamic drive toward equilibrium} \tag{1}$$

$$F^{+}: \quad \left.\frac{dF}{dt}\right|_{S} > 0 \qquad \textit{free-energy expenditure on organisation} \tag{2}$$

$$F^{auto}: \quad \ln P(\textit{state} \mid \textit{history}) \qquad \textit{log-predictive density from the past} \tag{3}$$

The intersection of F^{-} and F^{+} (equal and opposite pressure) defines the present *of the system. F^{auto} determines how strongly the past propagates into the future without active modulation.*

2.2 The Agency Operator F^{Φ}

Definition 2.2 (Agency Operator — Formal Self-Criterion)**.** *The agency operator F^{Φ} distinguishes self-determining systems from passive ones:*

$$F^{\Phi}(S,t) \;=\; D_{KL}\Big(P_{\textit{active-future}} \parallel P_{\textit{automatic-future}}\Big) \tag{4}$$

A system S possesses selfhood at time t if and only if $F^{\Phi}(S,t) > 0$. The degree of selfhood is monotonically related to F^{Φ}.

The selfhood continuum implied by Definition 2.2 is:

$F^{\Phi} = 0$	Stone, gas, random walk
$F^{\Phi} > 0$, small	Bacterium, reflex arc
$F^{\Phi} > 0$, moderate	Animal, institution
$F^{\Phi} > 0$, large	Reflective human, scientific community

2.3 Gap Dynamics

Definition 2.3 (Probability Gap)**.**

$$Gap(t) \;=\; H\Big(P_{future}(t)\Big) \;=\; -\sum_{s} p(s)\ln p(s) \tag{5}$$

$$\frac{d\,Gap}{dt} \;=\; \alpha\,F^{+} \;-\; \beta\,F^{-} \;-\; \gamma\,(1-\textit{self-active}) \tag{6}$$

The gap is large when the future is open (many equally probable paths) and collapses to zero at actualisation.

The temporal profile of Gap(t) follows a characteristic *ʔ-shape*: a long horizontal segment (Gap large, each observation barely narrows futures), a sharp corner at the critical point t^*, and a rapid post-t^* collapse. A strong F^{Φ} shifts t^* toward actualisation, preserving freedom of choice until the last moment.

2.4 Hierarchical Recursion

Proposition 2.1 (Hierarchical Recursion Principle)**.** ***[D]****The active modulation at level n becomes the automatic feedback at level $n+1$:*

$$F_n^{\Phi} \xrightarrow{\textit{stabilises}} F_{n+1}^{auto}$$

This explains why each emergent level has its own laws — they are the stabilised agency of the level below.

Transition	**Mechanism**
Physics → Chemistry	Quantum selection stabilises reaction pathways as F^{auto}
Chemistry → Life	Catalytic modulation stabilises metabolic networks
Life → Mind	Natural selection stabilises cognitive habits
Mind → Culture	Learning/reflection stabilises institutions
Culture → Civilisation	Science/art/law stabilises civilisational patterns

3 Ontological Layers

Definition 3.1 (Thermodynamic Flow Layer)**.** *A thermodynamic flow layer $\mathcal{L}_n$ is a stable attractor state of civilisational organisation characterised by: (i) a dominant flow operator F_n^+; (ii) a binding scarcity constraint; and (iii) a corresponding scarcity-to-abundance transition at time t_n.*

Every layer produces the scarcity that births the next: Energy built the machine that drowned us in information. Information built the agent that is now absorbing our will.

Table 2: The four ontological layers with their operators and saturation pathologies

Layer	Operator	Binding Constraint	Saturation Pathology	Window
$\mathcal{L}_E$ — Energy	F^+_{physical}	Muscle / mechanical work	Obesity, CVD, sedentary atrophy	∼100 yr
$\mathcal{L}_I$ — Information	$F^+_{\text{cognitive}}$	Memory / calculation	Depression, ADHD, radicalisation	∼55 yr
$\mathcal{L}_A$ — Agency	$F^+_{\text{volitional}}$	Decisions / goals	EDD, nihilism, identity diffusion	∼8 yr
$\mathcal{L}_\Phi$ — Meaning	$F^+_{\text{axiological}}$	Definitional framework	Not yet articulable	∼3 yr

3.1 The S-Curve Adoption Model

Each layer transition follows the logistic ODE:

$$\frac{dA}{dt} = k\,A\left(1 - \frac{A}{L}\right), \qquad A(t) = \frac{L}{1 + e^{-k(t-t_*)}} \tag{7}$$

where L is the saturation ceiling, k the diffusion rate, and t_* the inflection time (50% saturation). The adoption window between 10% and 90% saturation has a closed form:

$$\Delta t_{10\to 90} = \frac{2\ln 9}{k} \tag{8}$$

3.2 The Compression Law

Proposition 3.1 (Compression Law). *[C]*

$$\tau_n = \tau_0\, e^{-\alpha n}, \qquad \tau_0 \approx 71.1\, yr, \quad \alpha \approx 0.349 \tag{9}$$

Compression factor $e^{\alpha} \approx 1.42$ *per generation. Bootstrap 80% credible interval (agency layer,* $N = 50{,}000$ *resamples):* [2.6, 8.2] *years.*

3.3 The Fixed Point $T^* = 2043$

Empirically, layer-transition inflection times obey the linear map:

$$t_{n+1} = a\, t_n + b, \qquad a \approx 0.84, \quad b \approx 327.5, \quad r = 0.87\ (p = 0.012) \tag{10}$$

The fixed point is:

$$T^* = \frac{b}{1-a} \approx 2043\,\text{CE} \tag{11}$$

Because $a < 1$, T^* is a stable attractor. All layer transitions converge toward a single year. After T^*, the hierarchical recursion model makes no new predictions. [C]

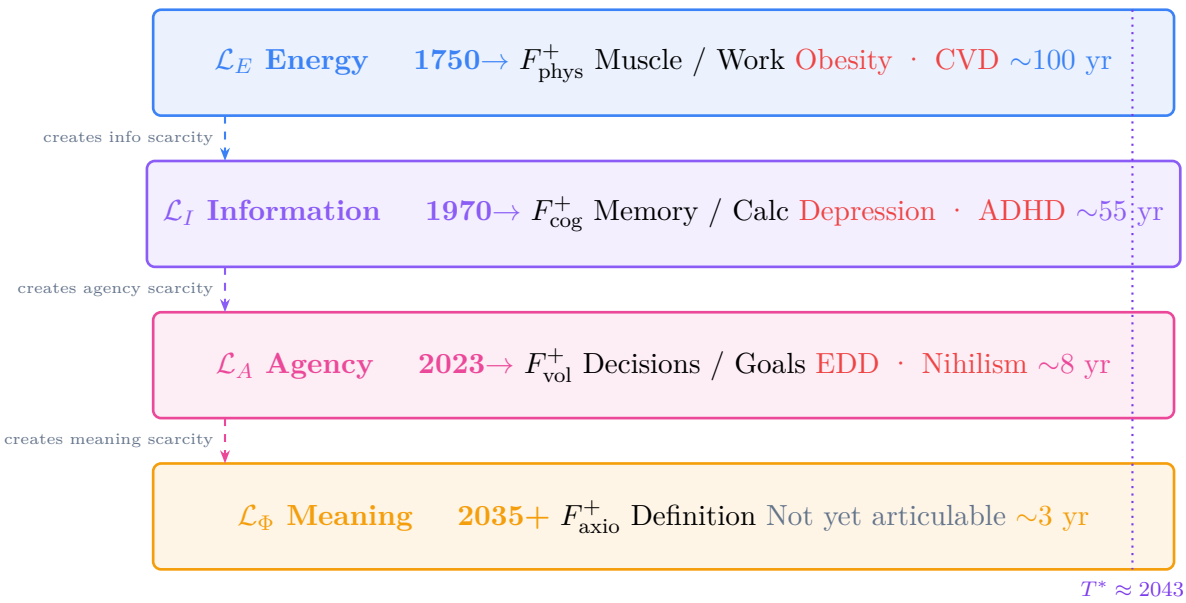

Figure 1: The four ontological layers. Each layer resolves one binding scarcity and generates the next. The adoption window shrinks by factor $e^{\alpha} \approx 1.42$ per layer (Compression Law). All layers converge toward $T^* \approx 2043$.

4 The Bifurcation Law

4.1 Formal Statement

Proposition 4.1 (Bifurcation Law). *[D]When a civilisational layer transition automates a previously scarce human capability, the population bifurcates into two stable attractors:*

$$F_n^+ \xrightarrow{revolution} \begin{cases} F_{auto}^- & \textit{Path A: delegation} \rightarrow \textit{atrophy} \\ F_{voluntary}^+ & \textit{Path B: voluntary practice} \rightarrow \textit{mastery} \end{cases} \tag{12}$$

Those who take Path B hold both the original capability and the new tool. Their combined output exceeds what any member of the pre-transition population could achieve. The bifurcation is not created by the technology — it is revealed by it.

4.2 The Gymnasium Recurrence Theorem

Theorem 4.2 (Gymnasium Recurrence). *[D]Every time civilisational abundance removes a category of necessary human effort, a compensatory institution emerges to reintroduce that effort as voluntary practice. The lag Δ between removal and institutional response satisfies:*

$$\Delta \propto \frac{1}{k_{adoption}} \tag{13}$$

Table 3: Historical instances of the Gymnasium Recurrence pattern

Ability Removed	Removing Technology	Compensatory Institution	Lag
Physical combat	City walls, 600 BCE	Gymnasium	∼200 yr
Manual transport	Steam engine, 1850	Modern gym	∼100 yr
Navigation skill	GPS, 2000 CE	Orienteering sport	∼25 yr
Cognitive agency	LLMs, 2023	Agency University (pred.)	5–10 yr [P]
Physical agency	Robotics, 2028	Embodied agency curriculum	3–7 yr [P]

4.3 Three Stable Attractors

Proposition 4.3 (Agency Stratification). *[C]In a fully automated economy, social stratification shifts from financial capital to Agency Capital C_A. Three stable attractors emerge:*

Attractor	***Condition***	***Dynamics***
Consumer class	$C_A \to 0$	*EDD onset; choice delegated; identity volatile*
Hybrid class	$C_A \in (0, C^*)$	*Unstable; oscillates between engagement and delegation*
Agency Elite	$C_A \geq C^*$	*Defines $\mathcal{L}_\Phi$; the Meaning layer*

Unlike financial inequality, Agency Capital inequality is largely self-selected *in the short run. The Consumer class is not forced into its position — it chooses comfort. The threat is that this choice becomes irreversible after t_{min}.*

5 Agency Capital and Epistemic Dependency Disorder

Definition 5.1 (Agency Capital). *Agency Capital C_A is the accumulated capacity of the F^Φ operator to generate Type III creative errors and to maintain purposive action in the absence of algorithmic recommendation. It is a stock variable with five canonical sources:*

1. ***Voluntary Friction:*** *repairing rather than replacing; physical discipline; intermittent fasting*
2. ***Cognitive Rebellion:*** *navigating without GPS; reading texts absent from LLM corpora*
3. ***Irrational Commitments:*** *ritual, long-term child-rearing, contemplative practice*
4. ***Somatic Epistemology:*** *extreme sports, craft mastery, musical performance*
5. ***Structured DMN Activation:*** *solitude, unstructured time, contemplative silence*

Definition 5.2 (Epistemic Dependency Disorder — EDD)**.** *EDD is a predicted clinical condition characterised by the inability to form stable preferences or initiate goal-directed action in the absence of algorithmic recommendation. It is the agency-layer analogue of Type 2 diabetes: as insulin receptors lose sensitivity to the hormone despite its abundance, EDD patients lose sensitivity to their own volitional signals despite their biological presence.*

Proposition 5.1 (EDD Onset Threshold)**.** ***[D]****EDD onset becomes mathematically inevitable when the Agency Capital atrophy rate exceeds its renewal rate. With representative parameters ($\delta = 0.15\, yr^{-1}$, $A_{max} = 1.0$, $A_{min} = 0.3$), the critical exercise threshold is:*

$$\varepsilon_{crit} = \frac{\delta(A_{max} - A_{min})}{A_{max}} \approx 0.064\, yr^{-1} \;\approx\; 23 \text{ days per year of } F^{\Phi}\text{-engaging activity} \tag{14}$$

Below this threshold, atrophy is irreversible.

Proposition 5.2 (Commodification Paradox)**.** ***[D]****When a source of Agency Capital is packaged as a commercial product, its F^{Φ}-generating potency decays exponentially:*

$$C(t) = C_0\, e^{-\lambda(t-t_c)}, \qquad \lambda > 0 \tag{15}$$

where t_c is the date of commodification. Agency Capital cannot be purchased — it can only be built. Buying is passive (F^{auto}). Building requires exactly the friction the market is designed to remove.

6 The Limits of AI

6.1 Four Independent Ceilings

Table 4: Four independent ceilings on AI capability

Ceiling	Formal Basis	Changeable?	Status
Landauer (thermodynamic)	$E_{\min} = k_B T \ln 2 \approx 3{\times}10^{-21}$ J / bit	No — absolute	[D]
Variance Collapse	$\sigma_n = \beta^n \sigma_0 \to 0$	No without ε	[D]
Corpus Saturation	Finite human-written text	Yes — delayed	[C]
Type III Error	Requires $F^\Phi > 0$; AI has $F^\Phi = 0$	Structural	[D]

6.2 The Variance-Collapse Theorem

Theorem 6.1 (Variance-Collapse Theorem). *[D]Let σ_n denote output-distribution variance after n generations of recursive AI-on-AI training:*

$$\sigma_{n+1} = \beta\,\sigma_n, \qquad 0 < \beta < 1 \implies \sigma_n = \beta^n\,\sigma_0 \xrightarrow{n\to\infty} 0 \tag{16}$$

A world of zero human creative error is the formal definition of frozen civilisational creativity.

The fix requires exogenous human variance injection per generation:

$$\sigma_{n+1} = \beta\,\sigma_n + \varepsilon, \qquad \varepsilon > 0 \implies \sigma^* = \frac{\varepsilon}{1-\beta} \tag{17}$$

This is the *thermodynamic value of error*: a non-zero human error rate is necessary, not merely tolerated, to maintain a non-degenerate representation of the world.

Corollary 6.2 (Civilisational Death State). *[D]When $H(t) \to 0$ (error entropy collapses), no new layer $\mathcal{L}_{n+1}$ can emerge because the generative material (F^Φ, Type III errors) is absent. The biological analogue: pre-antibiotic bacterial populations maintain high diversity at the cost of individual inefficiency. Perfect optimisation produces a monoculture maximally fragile to novel perturbations.*

7 Human–AI Symbiosis: The Paradigm Replacement

7.1 The Wrong System Boundary

Every existing benchmark, scaling law, and Landauer calculation treats AI as a system whose boundary ends at the hardware. This is the identical error made by a 19th-century anatomist who tried to evaluate the hunting capability of a domesticated wolf by measuring only the wolf's jaw — ignoring the human hand directing its pursuit.

The suppressed variable is the interaction term. When human and AI form a persistent interactive system, the agency operator becomes:

$$F^{\Phi}_{\text{hybrid}}(S,t) = F^{\Phi}_{\text{human}}(t) \;+\; F^{\Phi}_{\text{AI}}(t) \;+\; \underbrace{F^{\Phi}_{\text{interaction}}(t)}_{\text{unmeasured}} \tag{18}$$

The interaction term $F^{\Phi}_{\text{interaction}}(t)$ has no current measurement instrument, no SI unit, and no benchmark. Any claim about hybrid system capability that ignores this term is structurally incomplete. [X]

7.2 What Changes at the New System Boundary

Table 5: Limits under two system boundary assumptions

Limit	AI Alone	Human–AI Hybrid
Variance Collapse	$\sigma_n \to 0$, absolute	Resolved if $\varepsilon > 0$
Type III Error generation	Structurally impossible	Human injects into loop
Corpus saturation	Finite ceiling	Human generates continuously
Landauer thermodynamic	Absolute (physics)	Still absolute
Agency Capital erosion	N/A	**New critical limit:** if $C_A \to 0$, hybrid collapses

The new fundamental limit is not computational. It is volitional. The hybrid system's ceiling is determined by whether the human component maintains

$F^{\Phi}_{\text{human}} > F^{\Phi}_{\text{min}}$.

7.3 The Co-Evolution Ratio Problem

Proposition 7.1 (FM-III — Biological Discontinuity). *[C]Define the co-evolution ratio:*

$$R = \frac{r_T}{r_B}, \qquad t_{catch\text{-}up} = \frac{\ln R}{r_B} \tag{19}$$

where r_T is the technological adoption rate and r_B the biological adaptation rate. For agentic AI (2025–):

$$R_{AI} \approx 3{,}000, \qquad t_{catch\text{-}up} \approx 801 \text{ years} \tag{20}$$

Human biology requires 801 years to adapt to the deployment rate of agentic AI. This means FM-III (Biological Discontinuity) is active: the transition is mathematically possible but unexecutable by the biological substrate unless compensatory institutions are in place within the critical window.

8 The Critical Window: 2026–2031

Proposition 8.1 (Critical Mass Condition). *[C]Let $r_A(t)$ be the population fraction with $C_A \geq C_{min}$. A civilisational tipping point exists at r^*:*

$$\text{If } r_A(t_L) \geq r^* \;\Rightarrow\; \text{Renaissance scenario (self-sustaining)} \tag{21}$$

$$\text{If } r_A(t_L) < r^* \;\Rightarrow\; \text{Wall-E scenario (irreversible)} \tag{22}$$

Estimated $r^* = 15$–20%. Current $r_A(2026) \approx 8$–12%. The window is narrow. It is not closed.

Table 6: Two civilisational scenarios

Scenario	Defining Condition	Consequence
Wall-E	Institutions fail to adapt. F^{Φ} atrophy in the majority.	Decision-making concentrates in those who control agent infrastructure. C_A becomes the primary axis of inequality — and is irreversible.
Renaissance	Agency institutions constructed in parallel with AI deployment.	The majority maintains sufficient C_A to navigate the transition and collectively define $\mathcal{L}_{\Phi}$.

We are not building the future. We are deciding, right now in 2026, whether there will be anyone capable of building it. The answer is not in the hardware. The answer is in how many people wake up tomorrow and choose the harder path for no reason the algorithm would understand.

9 Falsifiable Predictions

All predictions carry explicit falsification conditions and time horizons. A prediction not testable within 20 years carries an [X] flag.

Table 7: Falsifiable predictions with time horizons and status

ID	Horizon	Prediction	Status
P1	2026–2028	The first major AI agent liability case remains unresolved due to absent legal frameworks.	[C]
P2	2027–2030	A novel clinical symptom cluster (preference-formation paralysis, goal-initiation failure, identity diffusion without algorithmic input) appears; initially misclassified as depression variant.	[C]
P3	2028–2032	The inflection t^* of agentic AI occurs: $\geq$ 50% of knowledge-work tasks in OECD economies initiated or completed by AI agent.	[C]
P4	2030–2035	Agency inequality ($\mathrm{Var}(C_A)$ across deciles) becomes measurable and statistically significant, distinct from income inequality.	[C]
P5	2026–2031	Institutions explicitly designed to build collective Agency Capital emerge; survival rate negatively correlated with commodification degree.	[P]
P6	2033–2040	AI systems trained exclusively on AI-generated content show $> 50\%$ reduction in distributional tail coverage after 10 generations, consistent with $\sigma_n \to 0$.	[D]
P7	2040–2050	$T^* \approx 2043$ CE observable as convergence of transition inflection times; no new layer identifiable by standard civilisational metrics thereafter.	[C]

Falsification criterion for P6 (strongest prediction): If AI systems trained exclusively on synthetic content do *not* show reduced tail coverage after 10 generations ($\sigma_n/\sigma_0 > 0.5$), the Variance-Collapse Theorem requires revision from its formal derivation.

10 Known Limitations

The following limitations are non-suppressible and propagate to all downstream claims:

L-1 The coupling constants α, β in Gap dynamics are fitted, not derived from first principles. Predictions are qualitative until formally derived.

L-2 The interaction term $F^{\Phi}_{\text{interaction}}(t)$ in Eq. (18) has no measurement instrument. Claims about hybrid system behaviour cannot exceed CONSISTENT epistemic status.

L-3 $T^* = 2043$ is conditioned on the historical compression pattern holding for future layers. If a layer does not follow logistic adoption dynamics, the fixed-point calculation fails.

L-4 The 15–20% critical mass threshold for the Renaissance/Wall-E bifurcation is estimated from historical precedent, not derived from first principles.

L-5 The EDD onset threshold $\varepsilon_{\text{crit}} \approx 0.064$ uses representative parameter values. Individual variation in δ, $A_{\max}$, and $A_{\min}$ will shift the threshold.

11 Epistemic Status Index

Tag	Meaning	Claims in this paper
[D]	Follows necessarily from axioms by logic or mathematics	Variance-Collapse Theorem; F^{Φ} selfhood criterion; Hierarchical Recursion; EDD onset threshold; P6
[C]	Plausible, motivated, falsifiable; awaiting confirmation	Compression Law coefficients; $T^* = 2043$; Three-Attractor distribution; FM-III activation; P1–P4, P7
[P]	Testable within declared timeframe	Gymnasium Recurrence predictions; P5
[X]	Beyond current formalism; no instrument exists	Content of $\mathcal{L}_{\Phi}$; value of $F^{\Phi}_{\text{interaction}}$; post-$T^*$ dynamics

12 Conclusion: We Have Been Here Before

The wolf that crossed the boundary between predator and partner did not do so because it understood the consequences. It did so because the short-term incentive

was unambiguous. So did the human. Neither species planned 40,000 years of co-evolution, a reorganisation of both genomes, and the construction of a hybrid cognitive-olfactory system more capable than either alone.

We are at an identical inflection. The short-term incentive to delegate F^{Φ} to an AI agent is as unambiguous as the proto-wolf's incentive to approach a human campfire. The question that was not available to the wolf — but is available to us — is: *what atrophies when we offload, and can we afford to lose it?*

The answer the framework delivers is precise: the Landauer ceiling cannot change; the Variance-Collapse theorem is irreversible without injection; the thermodynamic death state is formally reachable. But the injection mechanism — human Agency Capital maintained through voluntary friction — is available to any individual, independently of infrastructure or policy.

The bifurcation is not between humans and machines. It is between humans who maintain $F^{\Phi} > F^{\Phi}_{\min}$ and those who do not. The former will be, by every formal metric, more capable than any human in history. The latter will be more comfortable than any human in history, for as long as the infrastructure holds.

The window is narrow. It is not closed.

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